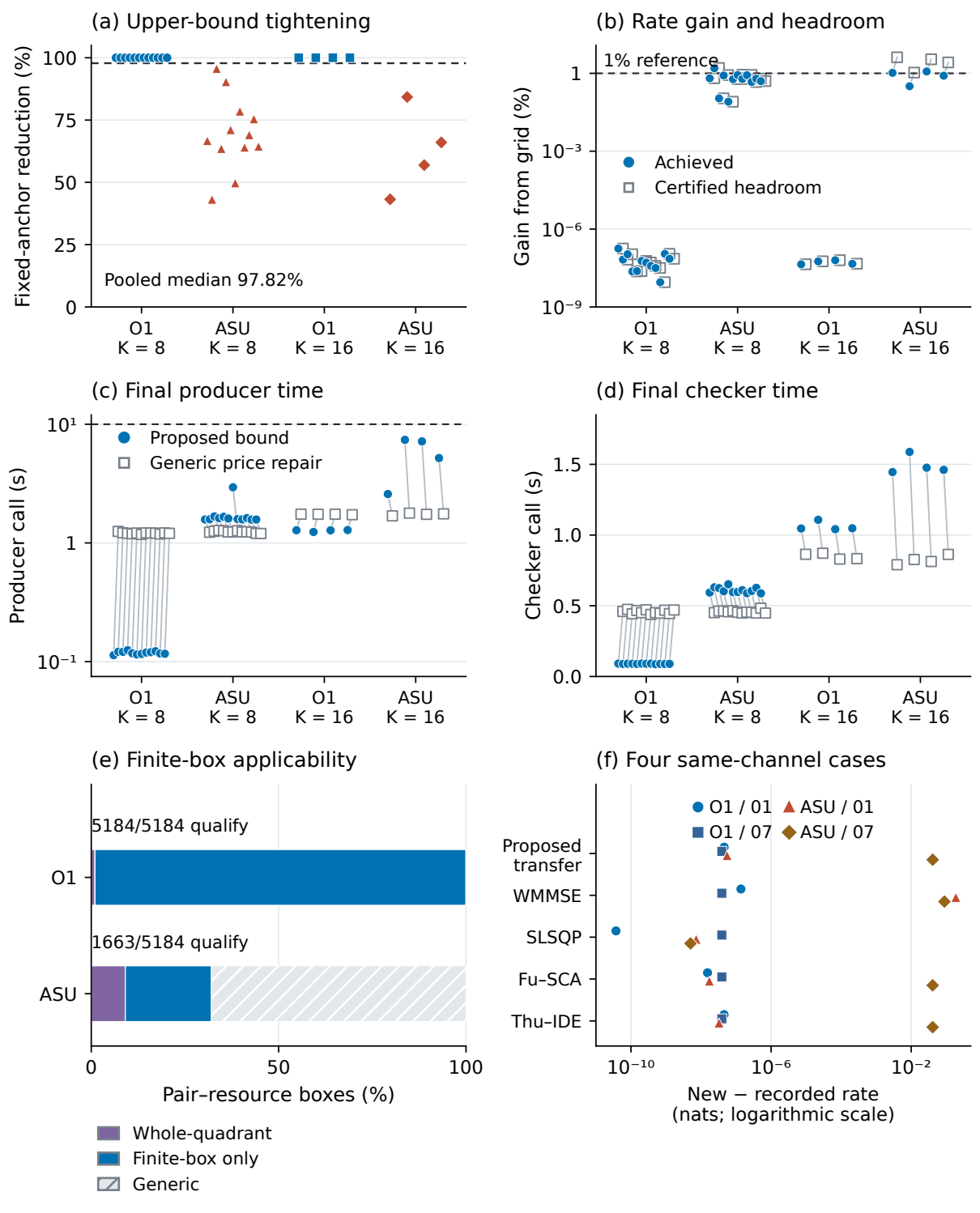
(a) Upper-bound tightening
Fixed-anchor reduction (%)
100
75
50
25
0
Pooled median 97.82%
O1
K = 8
ASU
K = 8
O1
K = 16
ASU
K = 16
(b) Rate gain and headroom
Gain from grid (%)
1% reference
1
10^{-3}
10^{-6}
10^{-9}
Achieved
Certified headroom
(c) Final producer time
Producer call (s)
10^{1}
1
10^{-1}
Proposed bound
Generic price repair
(d) Final checker time
Checker call (s)
1.5
1.0
0.5
0.0
(e) Finite-box applicability
5184/5184 qualify
O1
1663/5184 qualify
ASU
0
50
100
Pair–resource boxes (%)
Whole-quadrant
Finite-box only
Generic
(f) Four same-channel cases
O1 / 01
ASU / 01
O1 / 07
ASU / 07
Proposed transfer
WMMSE
SLSQP
Fu–SCA
Thu–IDE
10^{-10}
10^{-6}
10^{-2}
New − recorded rate
(nats; logarithmic scale)